

Sample Problem: Rural 2-lane Highways

LOS Table:

Table 16.4: Level of Service Criteria for Two-Lane Rural Highways

Level of Service	Class I Highways		Class II Highways PTSF (%)	Class III Highways PFFS (%)
	ATS (mi/h)	PTSF (%)		
A	>55	≤35	≤40	>91.7
B	>50–55	>35–50	>40–55	>83.3–91.7
C	>45–50	>50–65	>55–70	>75.0–83.3
D	>40–45	>65–80	>70–85	>66.7–75.0
E	≤40	>80	>85	≤66.7

(Source: *Two-Lane Highways*, Draft Chapter 15, National Cooperative Highway Research Program Project 3-92, Exhibit 15-3.)

Steps:

Average Travel Speed, ATS	Percent Time Spent Following (PTSF)
$ATS_d = FFS_d - 0.00776(\nu_d + \nu_o) - f_{npA}$ • ATS_d = ATS for direction d, mph • FFS_d = FFS for direction d, mph • ν_d = Demand Flow rate for direction d, pc/h • ν_o = Demand Flow rate for direction o opposite to direction d, pc/h • f_{npA} = adjustment for “No Passing” zones, % • $FFS_d = BFFS - f_{LS} - f_A$ ▪ $BFFS$ = base FFS for the facility, mph ▪ f_{LS} = adj. for lane and shoulder width, mph ▪ f_A = adj for access point density, mi/h • For direction d, Demand Flow Rate, ν_d, pc/h ▪ $\nu_d = \frac{V}{PHF \times f_{HV} \times f_G}$ ◦ V = hourly demand volume under prevailing conditions, veh/h ◦ PHF: peak hour factor ◦ f_{HV}: adjustment for heavy vehicle presence ◦ $f_{HV} = \frac{1}{1 + P_T(E_T - 1) + P_R(E_R - 1)}$ ◦ P_T = proportion of trucks and buses in the traffic stream ◦ P_R = proportion of recreational vehicles in the traffic stream ◦ E_T = passenger-car equivalent for trucks and buses ◦ E_R = passenger-car equivalent for recreational vehicles ◦ f_G: adjustment for grades	$PTSF_d = BPTSF_d + f_{npA} \left(\frac{\nu_d}{\nu_d + \nu_o} \right)$ • $PTSF_d$ = PTSF for direction d, % • $BPTSF_d$ = Base PTSF for direction d, % • ν_d = Demand Flow rate for direction d, pc/h • ν_o = Demand Flow rate for direction o opposite to direction d, pc/h • f_{npA} = adjustment for “No Passing” zones, % • $BPTSF_d$ = Base PTSF for direction d, % ▪ $BPTSF_d = 100[1 - e^{a\nu_d^b}]$ ◦ a, b = calibration constants based on opposing flow rate (o) in single-direction analysis

Sample Problem: Rural 2-lane Highways

Problem:

A Class I two-lane highway in rolling terrain has a peak demand volume of 500 veh/h, with 15% trucks and 5% RVs. The highway serves as a main link to a popular recreation area. The directional split of traffic is 60-40 during peak periods, and the peak-hour factor is 0.88. The 10-mile section under study has 40% “No Passing” zones. The base free-flow speed of the facility may be taken to be 60 mi/h. Lane widths are 12 feet, and shoulder widths are 2 feet. There are 10 access points per mile along this 10-mile section.

Sample Problem: Rural 2-lane Highways

Chapter 16: Two-Lane Highways

4th Ed: Tables

Table 16.5: Free-Flow Speed Adjustments for Lane and Shoulder Width

Lane Width (ft)	Reduction in FFS (mi/h), f_{LS}			
	Shoulder Width (ft)			
	$\geq 0 < 2$	$\geq 2 < 4$	$\geq 4 < 6$	≥ 6
$\geq 9 < 10$	6.4	4.8	3.5	2.2
$\geq 10 < 11$	5.3	3.7	2.4	1.1
$\geq 11 < 12$	4.7	3.0	1.7	0.4
≥ 12	4.2	2.6	1.3	0.0

(Source: Used by Permission of Transportation Research Board, National Science Council, *Highway Capacity Manual*, 4th Edition, Washington DC, 2000, Exhibit 20-5, p. 20-6.)

Table 16.6: Free-Flow Speed Adjustments for Access point Density

Access Points Per Mi	Reduction in FFS (mi/h) f_A
0	0.0
10	2.5
20	5.0
30	7.5
40	10.0

(Source: Used by Permission of Transportation Research Board, National Science Council, *Highway Capacity Manual*, 4th Edition, Washington DC, 2000, Exhibit 20-6, p. 20-6.)

Table 16.7: Grade Adjustment Factor (f_G) for General Terrain Segments and Specific Downgrades (ATS and PTSF Determinations)

One Direction Demand Flow Rate, $v = V/PHF$ (veh/h)	Level Terrain and Specific Downgrades, ATS and PTSF	Rolling Terrain ATS	Rolling Terrain PTSF
≤ 100	1.00	0.67	0.73
200	1.00	0.75	0.80
300	1.00	0.83	0.85
400	1.00	0.90	0.90
500	1.00	0.95	0.96
600	1.00	0.97	0.97
700	1.00	0.98	0.99
800	1.00	0.99	1.00
≥ 900	1.00	1.00	1.00

Note: Interpolation to the nearest 0.01 is recommended.

(Source: *Two-Lane Highways*, Draft Chapter 15, National Cooperative Highway Research Program Project 3-92, Exhibits 15-9 and 15-16.)

Table 16.8: Grade Adjustment Factor (f_G) for Specific Upgrades: ATS Determinations

Grade (%)	Grade Length (mi)	Directional Demand Flow Rate, v_{ph} (veh/h)								
		≤100	200	300	400	500	600	700	800	≥900
≥3 <3.5	0.25	0.78	0.84	0.87	0.91	1.00	1.00	1.00	1.00	1.00
	0.50	0.75	0.83	0.86	0.90	1.00	1.00	1.00	1.00	1.00
	0.75	0.73	0.81	0.85	0.89	1.00	1.00	1.00	1.00	1.00
	1.00	0.73	0.79	0.83	0.88	1.00	1.00	1.00	1.00	1.00
	1.50	0.73	0.79	0.83	0.87	0.99	0.99	1.00	1.00	1.00
	2.00	0.73	0.79	0.82	0.86	0.98	0.98	0.99	1.00	1.00
	3.00	0.73	0.78	0.82	0.85	0.95	0.96	0.96	0.97	0.98
	≥4.00	0.73	0.78	0.81	0.85	0.94	0.94	0.95	0.95	0.96
≥3.5 <4.5	0.25	0.75	0.83	0.86	0.90	1.00	1.00	1.00	1.00	1.00
	0.50	0.72	0.80	0.84	0.88	1.00	1.00	1.00	1.00	1.00
	0.75	0.67	0.77	0.81	0.86	1.00	1.00	1.00	1.00	1.00
	1.00	0.65	0.73	0.77	0.81	0.94	0.95	0.97	1.00	1.00
	1.50	0.63	0.72	0.76	0.80	0.93	0.95	0.96	1.00	1.00
	2.00	0.62	0.70	0.74	0.79	0.93	0.94	0.96	1.00	1.00
	3.00	0.61	0.69	0.74	0.78	0.92	0.93	0.94	0.98	1.00
	≥4.00	0.61	0.69	0.73	0.78	0.91	0.91	0.92	0.96	1.00
≥4.5 <5.5	0.25	0.71	0.79	0.83	0.88	1.00	1.00	1.00	1.00	1.00
	0.50	0.60	0.70	0.74	0.79	0.94	0.95	0.97	1.00	1.00
	0.75	0.55	0.65	0.70	0.75	0.91	0.93	0.95	1.00	1.00
	1.00	0.54	0.64	0.69	0.74	0.91	0.93	0.95	1.00	1.00
	1.50	0.52	0.62	0.67	0.72	0.88	0.90	0.93	1.00	1.00
	2.00	0.51	0.61	0.66	0.71	0.87	0.89	0.92	0.99	1.00
	3.00	0.51	0.61	0.65	0.70	0.86	0.88	0.91	0.98	0.99
	≥4.00	0.51	0.60	0.65	0.69	0.84	0.86	0.88	0.95	0.97
≥5.5 <6.5	0.25	0.57	0.68	0.72	0.77	0.93	0.94	0.96	1.00	1.00
	0.50	0.52	0.62	0.66	0.71	0.87	0.90	0.92	1.00	1.00
	0.75	0.49	0.57	0.62	0.68	0.85	0.88	0.90	1.00	1.00
	1.00	0.46	0.56	0.60	0.65	0.82	0.85	0.88	1.00	1.00
	1.50	0.44	0.54	0.59	0.64	0.81	0.84	0.87	0.98	1.00
	2.00	0.43	0.53	0.58	0.63	0.81	0.83	0.86	0.97	0.99
	3.00	0.41	0.51	0.56	0.61	0.79	0.82	0.85	0.97	0.99
	≥4.00	0.40	0.50	0.55	0.61	0.79	0.82	0.85	0.97	0.99
≥6.5	0.25	0.54	0.64	0.68	0.73	0.88	0.90	0.92	1.00	1.00
	0.50	0.43	0.53	0.57	0.62	0.79	0.82	0.85	0.98	1.00
	0.75	0.39	0.49	0.54	0.59	0.77	0.80	0.83	0.96	1.00
	1.00	0.37	0.45	0.50	0.54	0.74	0.77	0.81	0.96	1.00
	1.50	0.35	0.45	0.49	0.54	0.71	0.75	0.79	0.96	1.00
	2.00	0.34	0.44	0.48	0.53	0.71	0.74	0.78	0.94	0.99
	3.00	0.34	0.44	0.48	0.53	0.70	0.73	0.77	0.93	0.98
	≥4.00	0.33	0.43	0.47	0.52	0.70	0.73	0.77	0.91	0.95

Note: Straight-line interpolation of f_{GATS} for length of grade and demand flow permitted to the nearest 0.01.

(Source: Two-Lane Highways, Draft Chapter 15, National Cooperative Highway Research Program Project 3-92, Exhibit 15-10.)

Table 16.9: Grade Adjustment Factor (f_G) for Specific Upgrades: PTSF Determinations

Grade (%)	Grade Length (mi)	Directional Demand Flow Rate, $v = V/PHF$ (veh/h)								
		≤ 100	200	300	400	500	600	700	800	≥ 900
$\geq 3 < 3.5$	0.25	1.00	0.99	0.97	0.96	0.92	0.92	0.92	0.92	0.92
	0.50	1.00	0.99	0.98	0.97	0.93	0.93	0.93	0.93	0.93
	0.75	1.00	0.99	0.98	0.97	0.93	0.93	0.93	0.93	0.93
	1.00	1.00	0.99	0.98	0.97	0.93	0.93	0.93	0.93	0.93
	1.50	1.00	0.99	0.98	0.97	0.94	0.94	0.94	0.94	0.94
	2.00	1.00	0.99	0.98	0.98	0.95	0.95	0.95	0.95	0.95
	3.00	1.00	1.00	0.99	0.99	0.97	0.97	0.97	0.96	0.96
	≥ 4.00	1.00	1.00	1.00	1.00	1.00	0.99	0.99	0.97	0.97
$\geq 3.5 < 4.5$	0.25	1.00	0.99	0.98	0.97	0.94	0.93	0.93	0.92	0.92
	0.50	1.00	1.00	0.99	0.99	0.97	0.97	0.97	0.96	0.95
	0.75	1.00	1.00	0.99	0.99	0.97	0.97	0.97	0.96	0.96
	1.00	1.00	1.00	0.99	0.99	0.97	0.97	0.97	0.97	0.97
	1.50	1.00	1.00	0.99	0.99	0.97	0.97	0.97	0.97	0.97
	2.00	1.00	1.00	0.99	0.99	0.98	0.98	0.98	0.98	0.98
	3.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
	≥ 4.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
$\geq 4.5 < 5.5$	0.25	1.00	1.00	1.00	1.00	1.00	0.99	0.99	0.97	0.97
	≥ 0.50	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
≥ 5.5	All	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00

Note: Interpolation for length of grade and demand flow rate to the nearest 0.01 is recommended.

(Source: Two-Lane Highways, Draft Chapter 15, National Cooperative Highway Research Program Project 3-92, Exhibit 15-17.)

Table 16.10: Passenger Car Equivalents for General Terrain Segments: ATS and PTSF Determinations

Vehicle Type	Range of One-Way Flows, v (veh/h) $v = V/PHF$	For ATS Determination		For PTSF Determination	
		Level Terrain Specific Downgrades	Rolling Terrain	Level Terrain Specific Downgrades	Rolling Terrain
Trucks and Buses E_T	≤ 100	1.9	2.7	1.1	1.9
	200	1.5	2.3	1.1	1.8
	300	1.4	2.1	1.1	1.7
	400	1.3	2.0	1.1	1.6
	500	1.2	1.8	1.0	1.4
	600	1.1	1.7	1.0	1.2
	700	1.1	1.6	1.0	1.0
	800	1.1	1.4	1.0	1.0
	≥ 900	1.0	1.3	1.0	1.0
Recreational Vehicles E_R	All	1.0	1.1	1.0	1.0

(Source: Two-Lane Highways, Draft Chapter 15, National Cooperative Highway Research Program Project 3-92, Exhibits 15-10 and 15-18.)

Table 16.11: Passenger Car Equivalents for Trucks (E_T) on Specific Upgrades: ATS Determination

Grade (%)	Grade Length (mi)	Directional Demand Flow Rate, $v = V/PHF$ (veh/h)								
		≤100	200	300	400	500	600	700	800	≥900
≥3 <3.5	0.25	2.6	2.4	2.3	2.2	1.8	1.8	1.7	1.3	1.1
	0.50	3.7	3.4	3.3	3.2	2.7	2.6	2.6	2.3	2.0
	0.75	4.6	4.4	4.3	4.2	3.7	3.6	3.4	2.4	1.9
	1.00	5.2	5.0	4.9	4.9	4.4	4.2	4.1	3.0	1.6
	1.50	6.2	6.0	5.9	5.8	5.3	5.0	4.8	3.6	2.9
	2.00	7.3	6.9	6.7	6.5	5.7	5.5	5.3	4.1	3.5
	3.00	8.4	8.0	7.7	7.5	6.5	6.2	6.0	4.6	3.9
	≥4.00	9.4	8.8	8.6	8.3	7.2	6.9	6.6	4.8	3.7
≥3.5 <4.5	0.25	3.8	3.4	3.2	3.0	2.3	2.2	2.2	1.7	1.5
	0.50	5.5	5.3	5.1	5.0	4.4	4.2	4.0	2.8	2.2
	0.75	6.5	6.4	6.5	6.5	6.3	5.9	5.6	3.6	2.6
	1.00	7.9	7.6	7.4	7.3	6.7	6.6	6.4	5.3	4.7
	1.50	9.6	9.2	9.0	8.9	8.1	7.9	7.7	6.5	5.9
	2.00	10.3	10.1	10.0	9.9	9.4	9.1	8.9	7.4	6.7
	3.00	11.4	11.3	11.2	11.2	10.7	10.3	10.0	8.0	7.0
	≥4.00	12.4	12.2	12.2	12.1	11.5	11.2	10.8	8.6	7.5
≥4.5 <5.5	0.25	4.4	4.0	3.7	3.5	2.7	2.7	2.7	2.6	2.5
	0.50	6.0	6.0	6.0	6.0	5.9	5.7	5.6	4.6	4.2
	0.75	7.5	7.5	7.5	7.5	7.5	7.5	7.5	7.5	7.5
	1.00	9.2	9.2	9.1	9.1	9.0	9.0	9.0	8.9	8.8
	1.50	10.6	10.6	10.6	10.6	10.5	10.4	10.4	10.2	10.1
	2.00	11.8	11.8	11.8	11.8	11.6	11.6	11.5	11.1	10.9
	3.00	13.7	13.7	13.6	13.6	13.3	13.1	13.0	11.9	11.3
	≥4.00	15.3	15.3	15.2	15.2	14.6	14.2	13.8	11.3	10.0
≥5.5 <6.5	0.25	4.8	4.6	4.5	4.4	4.0	3.9	3.8	3.2	2.9
	0.50	7.2	7.2	7.2	7.2	7.2	7.2	7.2	7.2	7.2
	0.75	9.1	9.1	9.1	9.1	9.1	9.1	9.1	9.1	9.1
	1.00	10.3	10.3	10.3	10.3	10.3	10.3	10.3	10.2	10.1
	1.50	11.9	11.9	11.9	11.9	11.8	11.8	11.8	11.7	11.6
	2.00	12.8	12.8	12.8	12.8	12.7	12.7	12.7	12.6	12.5
	3.00	14.4	14.4	14.4	14.4	14.3	14.3	14.3	14.2	14.1
	≥4.00	15.4	15.4	15.3	15.3	15.2	15.1	15.1	14.9	14.8
>6.5	0.25	5.1	5.1	5.0	5.0	4.8	4.7	4.7	4.5	4.4
	0.50	7.8	7.8	7.8	7.8	7.8	7.8	7.8	7.8	7.8
	0.75	9.8	9.8	9.8	9.8	9.8	9.8	9.8	9.8	9.8
	1.00	10.4	10.4	10.4	10.4	10.4	10.4	10.4	10.3	10.2
	1.50	12.0	12.0	12.0	12.0	11.9	11.9	11.9	11.8	11.7
	2.00	12.9	12.9	12.9	12.9	12.8	12.8	12.8	12.7	12.6
	3.00	14.5	14.5	14.5	14.5	14.4	14.4	14.4	14.3	14.2
	≥4.00	15.4	15.4	15.4	15.4	15.3	15.3	15.3	15.2	15.1

Note: Straight line interpolation of E_T for length of grade permitted to nearest 0.1.

(Source: Two-Lane Highways, Draft Chapter 15, National Cooperative Highway Research Program Project 3-92, Exhibit 15-12.)

Table 16.12: Passenger Car Equivalents of RV's (E_R) for Specific Upgrades: ATS Determination

Grade (%)	Grade Length (mi)	Directional Demand Flow Rate, $v = V/PHF$ (veh/h)								
		≤ 100	200	300	400	500	600	700	800	≥ 900
$\geq 3 < 3.5$	≤ 0.25	1.1	1.1	1.1	1.0	1.0	1.0	1.0	1.0	1.0
	$> 0.25 \leq 0.75$	1.2	1.2	1.1	1.1	1.0	1.0	1.0	1.0	1.0
	$> 0.75 \leq 1.25$	1.3	1.2	1.2	1.1	1.0	1.0	1.0	1.0	1.0
	$> 1.25 \leq 2.25$	1.4	1.3	1.2	1.1	1.0	1.0	1.0	1.0	1.0
	> 2.25	1.5	1.4	1.3	1.2	1.0	1.0	1.0	1.0	1.0
$3.5 < 4.5$	≤ 0.75	1.3	1.2	1.2	1.1	1.0	1.0	1.0	1.0	1.0
	$> 0.75 \leq 3.50$	1.4	1.3	1.2	1.1	1.0	1.0	1.0	1.0	1.0
	≥ 3.50	1.5	1.4	1.3	1.2	1.0	1.0	1.0	1.0	1.0
$\geq 4.5 < 5.5$	≤ 2.50	1.5	1.4	1.3	1.2	1.0	1.0	1.0	1.0	1.0
	> 2.50	1.6	1.5	1.4	1.2	1.0	1.0	1.0	1.0	1.0
$\geq 5.5 < 6.5$	≤ 0.75	1.5	1.4	1.3	1.1	1.0	1.0	1.0	1.0	1.0
	$> 0.75 \leq 2.50$	1.6	1.5	1.4	1.2	1.0	1.0	1.0	1.0	1.0
	$> 2.50 \leq 3.50$	1.6	1.5	1.4	1.3	1.2	1.1	1.0	1.0	1.0
	> 3.50	1.6	1.6	1.6	1.5	1.5	1.4	1.3	1.2	1.1
≥ 6.5	≤ 2.50	1.6	1.5	1.4	1.2	1.0	1.0	1.0	1.0	1.0
	$> 2.50 \leq 3.50$	1.6	1.5	1.4	1.2	1.3	1.3	1.3	1.3	1.3
	> 3.50	1.6	1.6	1.6	1.5	1.5	1.5	1.4	1.4	1.4

Note: Interpolation in this table is not recommended.

(Source: *Two-Lane Highways*, Draft Chapter 15, National Cooperative Highway Research Program Project 3-92, Exhibit 15-13.)

Table 16.13: Passenger Car Equivalents of Trucks (E_T) and RVs (E_R) on Specific Upgrades: PTSF Determination

Grade (%)	Grade Length (mi)	Directional Demand Flow Rate, v_{ph} (veh/h)								
		≤100	200	300	400	500	600	700	800	≥900
		Passenger Car Equivalents for Trucks (E_T)								
≥3 <3.5	≤2.00	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
	3.00	1.5	1.3	1.3	1.2	1.0	1.0	1.0	1.0	1.0
	≤4.00	1.6	1.4	1.3	1.3	1.0	1.0	1.0	1.0	1.0
≥3.5 <4.5	≤1.00	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
	1.50	1.1	1.1	1.0	1.0	1.0	1.0	1.0	1.0	1.0
	2.00	1.6	1.3	1.0	1.0	1.0	1.0	1.0	1.0	1.0
	3.00	1.8	1.4	1.1	1.2	1.2	1.2	1.2	1.2	1.2
≥4.5 <5.5	≥4.00	2.1	1.9	1.8	1.7	1.4	1.4	1.4	1.4	1.4
	≤1.00	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
	1.50	1.1	1.1	1.1	1.2	1.2	1.2	1.2	1.2	1.2
	2.00	1.7	1.6	1.6	1.6	1.5	1.4	1.4	1.3	1.3
	3.00	2.4	2.2	2.2	2.1	1.9	1.8	1.8	1.7	1.7
≥5.5 <6.5	≥4.00	3.5	3.1	2.9	2.7	2.1	2.0	2.0	1.8	1.8
	≤0.75	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
	1.00	1.0	1.0	1.1	1.1	1.2	1.2	1.2	1.2	1.2

(Continued)

Table 16.13: Passenger Car Equivalents of Trucks (E_T) and RVs (E_R) on Specific Upgrades: PTSF Determination (Continued)

Grade (%)	Grade Length (mi)	Directional Demand Flow Rate, v_{ph} (veh/h)								
		≤100	200	300	400	500	600	700	800	≥900
		Passenger Car Equivalents for Trucks (E_T)								
≥6.5	1.50	1.5	1.5	1.5	1.6	1.6	1.6	1.6	1.6	1.6
	2.00	1.9	1.9	1.9	1.9	1.9	1.9	1.9	1.8	1.8
	3.00	3.4	3.2	3.0	2.9	2.4	2.3	2.3	1.9	1.9
	≥4.00	4.5	4.1	3.9	3.7	2.9	2.7	2.6	2.0	2.0
	≤0.50	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
	0.75	1.0	1.0	1.0	1.0	1.1	1.1	1.1	1.0	1.0
	1.00	1.3	1.3	1.3	1.4	1.4	1.5	1.5	1.4	1.4
	1.50	2.1	2.1	2.1	2.1	2.0	2.0	2.0	2.0	2.0
	2.00	2.9	2.8	2.7	2.7	2.4	2.4	2.3	2.3	2.3
	3.00	4.2	3.9	3.7	3.6	3.0	2.8	2.7	2.2	2.2
	≥4.00	5.0	4.6	4.4	4.2	3.3	3.1	2.9	2.7	2.5
		Passenger Car Equivalents for RVs (E_R)								
All	All	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0

Note: Interpolation for both length of grade and demand flow rate to the nearest 0.1 is recommended.

(Source: Two-Lane Highways, Draft Chapter 15, National Cooperative Highway Research Program Project 3-92, Exhibit 15-19.)

Table 16.14: Passenger Car Equivalents for Trucks Operating at Crawl Speeds on Specific Downgrades - ATS Determination

Difference Between FFS and Truck Crawl Speed (mi/h)	Directional Demand Flow Rate, $v = V/PHF$ (veh/h)								
	≤100	200	300	400	500	600	700	800	≥900
≤15	4.7	4.1	3.6	3.1	2.6	2.1	1.6	1.0	1.0
20	9.9	8.9	7.8	6.7	5.8	4.9	4.0	2.7	1.0
25	15.1	13.5	12.0	10.4	9.0	7.7	6.4	5.1	3.8
30	22.0	19.8	17.5	15.6	13.1	11.6	9.2	6.1	4.1
35	29.0	26.0	23.1	20.1	17.3	14.6	11.9	2.2	6.5
≥40	35.9	32.3	28.6	24.9	21.4	18.1	14.7	11.3	7.9

Note: interpolation of E_{TC} for speed difference and demand flow rate is permitted to nearest 0.1.

(Source: *Two-Lane Highways*, Draft Chapter 15, National Cooperative Highway Research Program Project 3-92, Exhibit 15-14.)

Table 16.15: Adjustment to *ATS* for %NPZ, Opposing Demand Flow, and *FFS*

Opposing Demand Flow Rate, v_o (pc/h)	Percent No Passing Zones				
	≤20	40	60	80	100
FFS = ≥ 65 mi/h					
≤100	1.1	2.2	2.8	3.0	3.1
200	2.2	3.3	3.9	4.0	4.2
400	1.6	2.3	2.7	2.8	2.9
600	1.4	1.5	1.7	1.9	2.0
800	0.7	1.0	1.2	1.4	1.5
1,000	0.6	0.8	1.1	1.1	1.2
1,200	0.6	0.8	0.9	1.0	1.1
1,400	0.6	0.7	0.9	0.9	0.9
≥1,600	0.6	0.7	0.7	0.7	0.8
FFS = 60 mi/h					
≤100	0.7	1.7	2.5	2.8	2.9
200	1.9	2.9	3.7	4.0	4.2
400	1.4	2.0	2.5	2.7	3.9
600	1.1	1.3	1.6	1.9	2.0
800	0.6	0.9	1.1	1.3	1.4
1,000	0.6	0.7	0.9	1.1	1.2
1,200	0.5	0.7	0.9	0.9	1.1
1,400	0.5	0.6	0.8	0.8	0.9
≥1,600	0.5	0.6	0.7	0.7	0.7
FFS = 55 mi/h					
≤100	0.5	1.2	2.2	2.6	2.7
200	1.5	2.4	3.5	3.9	4.1
400	1.3	1.9	2.4	2.7	2.8
600	0.9	1.1	1.6	1.8	1.9
800	0.5	0.7	1.1	1.2	1.4
1,000	0.5	0.6	0.8	0.9	1.1
1,200	0.5	0.6	0.7	0.9	1.0
1,400	0.5	0.6	0.7	0.7	0.9
≥1,600	0.5	0.6	0.6	0.6	0.7
FFS = 50 mi/h					
≤100	0.2	0.7	1.9	2.4	2.5
200	1.2	2.0	3.3	3.9	4.0
400	1.1	1.6	2.2	2.6	2.7
600	0.6	0.9	1.4	1.7	1.9
800	0.4	0.6	0.9	1.2	1.3

(Continued)

Table 16.15: Adjustment to *ATS* for %NPZ, Opposing Demand Flow, and *FFS* (Continued)

Opposing Demand Flow Rate, v_o (pc/h)	Percent No Passing Zones				
	≤ 20	40	60	80	100
FFS = 50 mi/h					
1,000	0.4	0.4	0.7	0.9	1.1
1,200	0.4	0.4	0.7	0.8	1.0
1,400	0.4	0.4	0.6	0.7	0.8
$\geq 1,600$	0.4	0.4	0.5	0.5	0.5
FFS ≤ 45 mi/h					
≤ 100	0.1	0.4	1.7	2.2	2.4
200	0.9	1.6	3.1	3.8	4.0
400	0.9	0.5	2.0	2.5	2.7
600	0.4	0.3	1.3	1.7	1.8
800	0.3	0.3	0.8	1.1	1.2
1,000	0.3	0.3	0.6	0.8	1.1
1,200	0.3	0.3	0.6	0.7	1.0
1,400	0.3	0.3	0.6	0.6	0.7
$\geq 1,600$	0.3	0.3	0.4	0.4	0.6

Note: Interpolation of $f_{p,ATS}$ for percent no-passing zones, demand flow rate, and FFS to the nearest 0.1 is recommended.

(Source: *Two-Lane Highways*, Draft Chapter 15, National Cooperative Highway Research Program Project 3-92, Exhibit 15-15.)

Table 16.16: Adjustment (f_{np}) for the Effect of "No Passing" Zones on PTSF

Total Two-Way Flow Rate, $v = v_d + v_o$ (pc/h)	Percent No-Passing Zones					
	0	20	40	60	80	100
Directional Split = 50/50						
≤200	9.0	29.2	43.4	49.4	51.0	52.6
400	16.2	41.0	54.2	61.6	63.8	65.8
600	15.8	38.2	47.8	53.2	55.2	56.8
800	15.8	33.8	40.4	44.0	44.8	46.6
1,400	12.8	20.0	23.8	26.2	27.4	28.6
2,000	10.0	13.6	15.8	17.4	18.2	18.8
2,600	5.5	7.7	8.7	9.5	10.1	10.3
3,200	3.3	4.7	5.1	5.5	5.7	6.1
Directional Split = 60-40						
≤200	11.0	30.6	41.0	51.2	52.3	53.5
400	14.6	36.1	44.8	53.4	55.0	56.3
600	14.8	36.9	44.0	51.1	52.8	54.6
800	13.6	28.2	33.4	38.6	39.9	41.3
1,400	11.8	18.9	22.1	25.4	26.4	27.3
2,000	9.1	13.5	15.6	16.0	16.8	17.3
2,600	5.9	7.7	8.6	9.6	10.0	10.2
Directional Split = 70/30						
≤200	9.9	28.1	38.0	47.8	48.5	49.0
400	10.6	30.3	38.6	46.7	47.7	48.8
600	10.9	30.9	37.5	43.9	45.4	47.0
800	10.3	23.6	28.4	33.3	34.5	35.5
1,400	8.0	14.6	17.7	20.8	21.6	22.3
2,000	7.3	9.7	12.7	13.3	14.0	14.5
Directional Split = 80/20						
≤200	8.9	27.1	37.1	47.0	47.4	47.9
400	6.6	26.1	34.5	42.7	43.5	44.1
600	4.0	24.5	31.3	38.1	39.1	40.0
800	3.8	18.5	23.5	28.4	29.1	29.9
1,400	3.5	10.3	13.3	16.3	16.9	32.2
2,000	3.5	7.0	8.5	10.1	10.4	10.7
Directional Split = 90/10						
≤200	4.6	24.1	33.6	43.1	43.4	43.6
400	0.0	20.2	28.3	36.3	36.7	37.0
600	-3.1	16.8	23.5	30.1	30.6	31.1
800	-2.8	10.5	15.2	19.9	20.3	20.8
1,400	-1.2	5.5	8.3	11.0	11.5	11.9

Note: Straight-line interpolation of f_{np} for % "No Passing" zones, demand flow rate, and directional split is permitted to the nearest 0.1.

Table 16.17: Coefficients "a" and "b" for Use in Equation 16-11

Opposing Demand Flow Rate, v_o (pc/h)	Coefficient a	Coefficient b
≤ 200	-0.0014	0.973
400	-0.0022	0.923
600	-0.0033	0.870
800	-0.0045	0.833
1,000	-0.0049	0.829
1,200	-0.0054	0.825
1,400	-0.0058	0.821
$\geq 1,600$	-0.0062	0.817

Note: Straight-line interpolation of "a" and "b" for opposing demand flow is permitted to the nearest 0.001.

(Source: *Two-Lane Highways*, Draft Chapter 15, National Cooperative Highway Research Program Project 3-92, Exhibit 15-20.)

Table 16.18: Values Used in Converting Volumes to Flow Rates Under Equivalent Base Conditions

Factor	Source Table	Directional Demand Volume	
		$V_1 = 341 \text{ veh/h}$	$V_2 = 227 \text{ veh/h}$
$f_G(ATS)$	Table 16.7 (Rolling Terrain)	0.86	0.77
$E_T(ATS)$	Table 16.10 (Rolling Terrain)	2.1	2.3
$E_R(ATS)$	Table 16.10 (Rolling Terrain)	1.1	1.1
$f_G(PTSF)$	Table 16.7 (Rolling Terrain)	0.87	0.81
$E_T(PTSF)$	Table 16.10 (Rolling Terrain)	1.7	1.8
$E_R(PTSF)$	Table 16.10 (Rolling Terrain)	1.0	1.0

Table 16.20: Optimal Lengths of Passing Lanes on Two-Lane Highways

Directional Flow Rate (pc/h)	Optimal Length of Passing Lane (mi)
100	≤ 0.50
200	$> 0.50-0.75$
400	$> 0.75-1.00$
≥ 700	$> 1.00-2.00$

(Source: Used with permission of Transportation Research Board, National Research Council, *Highway Capacity Manual*, 4th Edition, Washington DC, 2000, Exhibit 12-12, p. 12-18).

Table 16.21: Effective Downstream Length of Passing Lanes

Directional Flow Rate (pc/h)	Downstream Length of Roadway Affected, L_{de} (mi)	
	ATS	PTSF
≤ 200	1.7	13.0
400	1.7	8.1
700	1.7	5.7
$\leq 1,000$	1.7	3.6

Note: Straight-line interpolation for L_{de} for directional flow rate is permitted to the nearest 0.1 mile.

(Source: Used with permission of Transportation Research Board, National Research Council, *Highway Capacity Manual*, 4th Edition, Washington DC, 2000, Exhibit 20-23, p. 20-24).

Table 16.22: Adjustment Factors for Passing Lanes in Single-Direction Segments of Two-Lane Highways

Directional Demand Flow Rate (pc/h)	Adjustment Factor, f_{pl} For ATS	Adjustment Factor, f_{pl} for PTSF
0-300	1.08	0.58
>300-600	1.10	0.61
>600	1.11	0.62

(Source: Used with permission of Transportation Research Board, National Research Council, *Highway Capacity Manual*, 4th Edition, Washington DC, 2000, Exhibit 20-24, p. 20-26.)

Table 16.23: Adjustment Factors for Climbing Lanes in Single-Direction Segments of Two-Lane Highways

Directional Demand Flow Rate (pc/h)	Adjustment Factor, f_{pl} for ATS	Adjustment Factor, f_{pl} for PTSF
0-300	1.02	0.20
>300-600	1.07	0.21
>600	1.14	0.23

(Source: Used with permission of Transportation Research Board, National Research Council, *Highway Capacity Manual*, 4th Edition, Washington DC, 2000, Exhibit 20-27, p. 20-29.)